AEGIS: Matrix-Free Diagnostics for the Adversarial Fault Lines of Graph Neural Networks

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Abstract

Graph neural networks now inform safety-critical decisions in drug-interaction screening, fraud detection, and power-grid contingency analysis, where adversarial perturbations of the graph can flip predictions. Deploying such a model demands two things current tools separate: a map of its *adversarial fault lines*, and a usable robustness *guarantee*. AEGISsupplies both from one matrix-free object, the *constrained sensitivity matrix* S_c, computed through a Neumann-series resolvent and randomized SVD that scale to N=7,650 on one GPU. From S_c we obtain (i) three first-order diagnostics—the SVD-optimal attack direction, per-edge sensitivity rankings, and per-node radii; (ii) AEGIS-*Conformal*, a distribution-free certificate whose worst-case conformity-score shift over the ε-ball is bounded *analytically* by S_c (no Monte-Carlo smoothing), giving $\Pr[y_v \in C_\varepsilon(v)] \geq 1-\alpha$ over the entire ball—coverage holds at the nominal level under the very worst-case attack it certifies (10 seeds, Cora and Citeseer; sets of $\sim$1.0–1.5 labels), *non-vacuous where deterministic radii are thin*; and (iii) a constant-factor two-sided characterisation of the structural-robustness boundary with critical budget $\varepsilon_{\mathrm{crit}}=(1-\kappa)/\left\|W\right\|_2$, showing the norm certificate is **2–9**$\times$ conservative relative to the spectral boundary an attack actually reaches. Regularizing $\sigma_1(S_c)$ improves robustness, so one operator drives attack, certification, and defense. Across 6 datasets, 7 architectures, and 4 domains (390 runs), the continuous-to-discrete bridge is positive in *all* $\mathbf{39/39}$ cells (median $\tau=+\mathbf{0.99}$, $p<10^{-5}$; $+\mathbf{0.996}$ on Amazon Photo, N=7,650), and one query delivers $\mathbf{74}$–$\mathbf{156}\times$ the per-query damage of 50-step PGD.

1 Introduction

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection via perturbed transaction edges (Zügner, Akbarnejad, and Günnemann 2018), drug-interaction models misfire under perturbed molecular graphs (Dai et al. 2018), and power grids can miss critical contingencies when their graph representations are structurally fragile (Nakiganda et al. 2023). A practitioner deploying a GNN faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Each existing thread maps only part of these *adversarial fault lines*. Structural attacks (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a; Geisler et al. 2021; Wu et al. 2019) rank edges by surrogate gradients but yield no continuous direction and no per-node budget; smoothing (Bojchevski, Klicpera, and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023, 2021) and deterministic certifiers (Li and Wang 2025; Zügner and Günnemann 2019b) return per-node certificates but no edge ranking or direction; robust-architecture defenses (Zhu et al. 2019; Zhang and Zitnik 2020; Jin et al. 2020) harden models without surfacing per-edge vulnerability; and sober-look critiques (Mujkanovic et al. 2022; Gosch et al. 2023) flag evaluation pitfalls. No prior method audits, certifies, and defends from a single matrix-free pass. AEGIS(Adversarial Evaluation of Graph Integrity via Sensitivity) does, from one analytical object: the constrained sensitivity matrix S_c (section 4). Intuitively, S_c records how strongly each edge can push the model's equilibrium prediction, so its leading singular direction, its column norms, and its per-node margins yield the three diagnostics in a single pass; the same analytic bound turns the per-node radius into a sound, distribution-free certificate (section 3.1). Here "one query" means one matrix-free S_c construction (a single randomized SVD over the equilibrium resolvent), not three separate analyses. The closed-form regime characterisation (Theorem 1) is established for contractive implicit GNNs, and S_c extends to any GNN with continuous edge-weight-modulated message passing (section 5.3); the threat model targets edge deletion and weight perturbation (section 2).

Contributions. (1) Constrained sensitivity operator. S_c specialises equilibrium IFT sensitivity (Koh and Liang 2017; Gould, Hartley, and Campbell 2021) to *structural* edge perturbations via the projection P_c. One matrix-free query reads the SVD-optimal direction, per-edge rankings, and per-node radii together, where no prior object yields all three; it matches a dense σ_1 to 0.03% at N=200 and scales to N=7,650. **(2) Certification from sensitivity.** AEGIS-Conformal turns the S_c bound into a *distribution-free* certificate—coverage $\geq 1-\alpha$ over the ε-ball, holding at the nominal level under worst-case attack and non-vacuous where the deterministic radius is thin (section 3.1); and a constant-factor two-sided characterisation of the boundary ($\varepsilon_{\mathrm{crit}}=(1-\kappa)/\left\|W\right\|_2$) shows norm certificates are 2–9$\times$ conservative relative to the spectral boundary an attack reaches (Theorem 1 and section D.3). **(3) Coupled

defense. Regularizing $\sigma_1(S_c)$ improves robustness, so the operator that finds the worst attack also drives the defense (section 3.2). **(4) Empirical evaluation.** 6 datasets, 7 architectures, 4 domains (390 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD (Geisler et al. 2021) with complementary positioning against AGNNCert (Li and Wang 2025) (section D.2), continuous-to-discrete transfer (fig. 3), and a fraud-detector audit (section 6).

2 Preliminaries and Threat Model

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes, adjacency $A \in \mathbb{R}^{N\times N}$, features $X \in \mathbb{R}^{N\times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$. Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i-th singular value, $\text{vec}(\cdot)$ column-major vectorization. The sensitivity matrices (S, constrained S_c, unrolled S_K, per-node block S_v) and their governing scalars (contractivity $\kappa{=}\|J_z\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\text{crit}}$, radius r_v) are defined where they first appear.

IGNN and IFT. AEGISanalyses models that settle to an equilibrium, because that fixed point is what exposes structural sensitivity in closed form. A DEQ-GNN (Bai, Kolter, and Koltun 2019; Bai, Koltun, and Kolter 2020) iterates a weight-tied F_θ to a fixed point; the IGNN instantiation (Gu et al. 2020) uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \tag{1}$$

with $W \in \mathbb{R}^{d\times d}$ spectral-normalized (Miyato et al. 2018) so $\kappa \leq \left\|\hat{A}\right\|_2 \|W\|_2 < 1$, ReLU ϕ, and a learned projection X_{proj}. Training uses implicit differentiation through the fixed point (Gould, Hartley, and Campbell 2021; Fung et al. 2022) with optional Jacobian regularization (Bai, Koltun, and Kolter 2021), and a linear head $f(z^*) = W_{\text{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A){=}z^*{-}F(z^*, A){=}0$ defines z^* as an implicit function of A, with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1}\frac{\partial F}{\partial p}\bigg|_{z^*}. \tag{2}$$

Equation (2) is the lever AEGISpulls: it gives the closed-form first-order change in z^* under any parameter p, with the resolvent $(I - J_z)^{-1}$ amplifying a perturbation by at most $1/(1{-}\kappa)$ via its convergent Neumann series. The pseudospectral index $\eta{=}\left\|(I - J_z)^{-1}\right\|_2 (1 - \rho) \geq 1$ (Trefethen and Embree 2005) measures the gap to the spectral-radius bound ($\eta{=}1$ for normal J_z). The IFT has served hyperparameter optimization (Lorraine, Vicol, and Duvenaud 2020) and implicit differentiation (Gould, Hartley, and Campbell 2021), but not adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \left\|\delta\hat{A}\right\|_F \leq \varepsilon, \tag{3}$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A}{=}\delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij}{=}0$ for $(i, j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIStargets edge-deletion / weight-perturbation; insertion attacks (Nettack-class (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a) and node-injection (Sun et al. 2020)) require enlarging the basis to $\bar{E} \supseteq E$, a natural follow-up that preserves the S_c construction. Discrete deletions connect to the continuous model through Proposition 3. Unlike input-feature perturbation (El Ghaoui et al. 2021; Revay, Wang, and Manchester 2020), structural perturbation changes the operator itself: z^* depends on A through both $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F/\partial\text{vec}(A)$, motivating S_c in section 3.

3 Adversarial Equilibrium Theory

Building on the threat model of section 2, we derive the machinery behind the three diagnostics of section 1 (per-edge rankings, the SVD-optimal direction, per-node radii) from one unified object, the constrained sensitivity matrix S_c. The governing picture is a phase transition in the budget ε: while ε is small the perturbed model stays contractive and its prediction moves boundedly with the graph; as ε approaches a critical budget $\varepsilon_{\text{crit}}$ the equilibrium grows arbitrarily sensitive; past $\varepsilon_{\text{crit}}$ contraction is no longer guaranteed and the prediction can break. Theorem 1 makes this precise and puts $\varepsilon_{\text{crit}}$ in closed form.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- *(A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT (Bolte and Pauwels 2021) (proof below).*
- *(A2) $\|W\|_2 \leq c$ via spectral normalization.*
- *(A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training (κ=0.14–0.59 across our suite; table 2). (A2) does not imply (A3); we report the trained κ alongside $\varepsilon_{\text{crit}}$ so practitioners can audit, and* AEGIS*still returns S_c rankings and SVD-optimal attacks if (A3) fails, the formal guarantees then scoped to the contractive regime.*

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \tag{4}$$

Then structural perturbations $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical *($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{5}$$

where $S = (I - J_z)^{-1}J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} /(1 - \kappa)$.

(b) Critical *($\varepsilon \to \varepsilon_{\text{crit}}$): The resolvent obeys $\left\|(I{-}J_z')^{-1}\right\|_2 \geq 1/\min_i |1{-}\lambda_i(J_z')|$, blowing up when an eigenvalue of J_z' approaches 1 (a vanishing spectral gap to 1, not $\|J_z'\|_2 \to 1$ alone). The rate is $\Omega(1/(\varepsilon_{\text{crit}}{-}\varepsilon))$ when J_z' is normal with a dominant real-positive (Perron) eigenvalue; in general $\varepsilon_{\text{crit}}$* lower-bounds *the divergence*

threshold, the slack being the nonnormality η (Observation 1; $\eta \leq 2.47$ for general ReLU, Remark 1).
(c) Supercritical *($\varepsilon \geq \varepsilon_{\text{crit}}$): The contraction certificate no longer applies ($\|J_z'\|_2$ may exceed 1); at the worst-case $\kappa = \left\|\hat{A}\right\|_2 \|W\|_2$, $\varepsilon < \varepsilon_{\text{crit}}$ is a sufficient safety boundary that trained IGNNs sit well inside (a 2–4× spectral-radius margin to ρ=1, section D.3—distinct from the 2–9×* certificate *conservatism of Theorem 2).*

Proof in section A.1. The subcritical bound follows from a conservative IFT (Bolte and Pauwels 2021) on each ReLU region with Neumann $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$; the critical rate from the unconditional resolvent identity $\left\|(I - M)^{-1}\right\|_2 \geq 1/\min_i |1 - \lambda_i(M)|$, which $\varepsilon_{\text{crit}}$ lower-bounds (slack η).

$\varepsilon_{\text{crit}}$ is a closed-form sufficient bound that S_c tightens 7–14× over the unconstrained $\sqrt{r}$ bound. The Neumann bound uses κ, not $\rho(J_z)$; the gap is the pseudospectral index η, bounded by W alone:

Observation 1 (Graph-Independent Nonnormality Bound, all-active case)**.** *For $J_z = \operatorname{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$ and all activations positive ($\phi' \equiv 1$): $\eta \leq \kappa(V_W)$ where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \left\|V_W^{-1}\right\|_2$, independent of $\hat{A}$.*

Proof in section A.2 (Kronecker eigenstructure of $J_z = \hat{A} \otimes W$ plus the diagonalisable resolvent bound).

Remark 1 (Empirical extension to general ReLU patterns)**.** *The proof above requires $\phi' \equiv 1$; for general ReLU patterns, $\eta \in [1.19, 2.47]$ on our suite (section D.4), tracking $\kappa(V_W)$ closely. Graph structure does not amplify nonnormality; the moderate η traces to spectral-norm regularisation of W.*

Theorem 1(a) certifies safety *below* $\varepsilon_{\text{crit}}$; the same operator also bounds the boundary from *above*. Let $\varepsilon^\star = 1/\rho(W) - \rho(\hat{A})$ be the all-active *spectral* break budget and $g_W = \|W\|_2 / \rho(W) \geq 1$ the nonnormality of W, which—for symmetric $\hat{A}$—is the *entire* norm-vs-radius gap (Observation 1).

Theorem 2 (Constant-factor two-sided boundary, all-active)**.** *Under* (A1)–(A3) *with $\hat{A}$ symmetric and $\varepsilon_{\text{crit}} > 0$, the all-active contraction boundary ε_{br} (the smallest feasible $\left\|\delta\hat{A}\right\|_F$ at which $\rho(\hat{A}' \otimes W) \geq 1$) obeys*

$$\varepsilon_{\text{crit}} \leq \varepsilon_{\text{br}} \leq \tfrac{C}{\beta}\,\varepsilon_{\text{crit}}, \qquad C = g_W\,\tfrac{1+\kappa}{1-\kappa}, \tag{6}$$

with C dimension-free and computable post-training, and $\beta \in (0, 1]$ the edge-supported Frobenius mass of $\hat{A}$'s Perron mode (positive for any connected graph). The upper side is attained by a rank-one critical-driving attack aligned with that mode; the bracket collapses to equality iff g_W=1 and β=1.

This is a *constant-factor* two-sided characterisation, not a tight one—the slack C/β reaches ∼10 on our suite. Its value is the coupling it exposes: one operator S_c supplies the certified budget $\varepsilon_{\text{crit}}$, the matching attack, and (through $\sigma_1(S_c)$) the defense of section 3.2. Empirically the norm certificate under-states the true breaking budget by **2–9**× across 10 seeds (resolvent divergence γ=1.02); the proof and nonlinear extension are in section B.

We now read the three diagnostics off S_c, beginning with the most damaging perturbation direction.

Proposition 1 (Maximally Sensitive First-Order Structural Perturbation)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot$ $\operatorname{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S, with shift $\varepsilon \cdot \sigma_1(S)$. Restricted to the threat model's feasible set (symmetric, edge-supported; section 2) the maximizer is instead the leading right singular vector of the constrained S_c (below) in the edge basis $\{b_k\}$, with shift $\varepsilon\,\sigma_1(S_c) \leq \varepsilon\,\sigma_1(S)$; the unconstrained* $\operatorname{reshape}(v_1)$ *is generally infeasible.*

The direction (Proposition 1) is optimal for hidden-state damage within the pseudospectral envelope of Observation 1 and Remark 1 and empirically aligns with classification damage (section 5.2). The *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with edge basis $b_k = (e_i e_j^\top + e_j e_i^\top)/\sqrt{2}$ ($\|b_k\|_F$ =1, so $\sigma_1(S_c)$ is consistent with $\left\|\delta\hat{A}\right\|_F = \varepsilon$); the $N^2 \to |E|$ projection P_c is the standard duplication-matrix reduction (Magnus and Neudecker 2019) on the edge-supported symmetric subspace, queried matrix-free in algorithm 1. This first-order envelope identifies the dangerous direction, not a global safety margin (section 5.1).

Proposition 2 (Per-Node First-Order Sensitivity Radius)**.** *For a linear head $f(z) = Wz + b$ and a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v}(z_v^*) - f_c(z_v^*) > 0$ to every competitor $c \neq y_v$, the first-order sensitivity radius*

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c)\,S_v\|_2} \tag{7}$$

preserves the classification of v for any $\left\|\delta\hat{A}\right\|_F < r_v$, where S_v are the block-rows of S at node v; substituting $S_{c,v}$ gives the tighter constrained radius $r_v^{(c)} \geq r_v$. Minimizing over all *competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes; the cheaper runner-up surrogate $\hat{r}_v$ can be optimistic when a non-runner-up class is more aligned.*

Proof in section A.3. A Cauchy–Schwarz bound on each per-competitor margin change gives r_v as the smallest margin divided by two amplification factors (decision-boundary and equilibrium sensitivity); high-margin, low-sensitivity nodes get large radii.

3.1 AEGIS-Conformal: A Distribution-Free Certificate

The radius r_v (eq. (7)) is a first-order threshold—locally tight, but violable at larger ε. We upgrade it to a *sound, distribution-free* guarantee. Split conformal prediction yields a set $C(v)$ with $\Pr[y_v \in C(v)] \geq 1 - \alpha$; making this hold *robustly* over the ball $\left\|\delta\hat{A}\right\|_F \leq \varepsilon$

Table 1: AEGIS-Conformal coverage (10 seeds; α=0.1, target 0.90). **Gate**: coverage under the worst-case attack at ε. "set": mean prediction-set size (of 7 Cora / 6 Citeseer classes). Std over seeds $\leq$0.06.

Data	score	ε	cov / set	gate
Cora	APS	0.01	0.901 / 1.37	**0.900**
	APS	0.05	0.892 / 1.06	**0.983**
	TPS	0.01	0.882 / 0.95	**0.895**
	TPS	0.05	0.893 / 0.99	**0.983**
Citeseer	APS	0.01	0.919 / 1.50	**0.925**
	APS	0.05	0.915 / 1.35	**0.968**
	TPS	0.01	0.910 / 1.21	**0.918**
	TPS	0.05	0.917 / 1.26	**0.957**

requires bounding the worst-case conformity-score shift, which AEGISsupplies *analytically* via the same S_c sensitivity (a curvature-corrected $L_1^c\varepsilon+C_v\varepsilon^2$), replacing the $\sim10^4$-sample smoothing otherwise needed. Feeding it into the binary-certificate split conformal of Zargarbashi, Antonelli, and Bojchevski (2023) yields a robust set with

$$\Pr\big[y_v \in C_\varepsilon(v)\big] \;\geq\; 1-\alpha \qquad \text{for all } \left\|\delta\hat{A}\right\|_F \leq \varepsilon. \quad (8)$$

Table 1 reports coverage over the 10 seeds. The *gate*—coverage under the worst-case AEGISattack at magnitude ε (reconverging the equilibrium after the v_1 perturbation)—holds at the nominal $1-\alpha$=0.90 at ε=0.01 and is conservative (0.92–0.98) at ε=0.05, with *zero* equilibrium divergence across all gate nodes; sets are $\sim$1.0–1.5 labels. The certificate is thus non-vacuous exactly where the deterministic radius certifies few nodes, uses zero Monte-Carlo samples, and via S_K applies to any GNN. Exchangeability on a single transductive graph is a stated condition (the empirical gate is the evidence); we form S_c densely at N=200, a matrix-free conformal pass being future work.

3.2 Coupling: A $\sigma_1(S_c)$ Defense

The operator that finds the worst attack also defends against it. Adding $\lambda\,\sigma_1(S_c)$ to the training loss shrinks the dominant sensitivity: at λ=3$\times$10^{-4} the IGNN reaches $\sigma_1(S_c)$=32.6$\pm$1.7 at 73.9$\pm$0.8% accuracy, raising certified coverage to 0.82$\pm$0.03. Across seeds the per-node attack magnitude and certified radius are partially anticorrelated ($-$0.65$\pm$0.12, margin-controlled), so suppressing the attack direction *is* the defense.

3.3 Continuous-to-Discrete Transfer

Proposition 3 (Continuous-to-Discrete Ranking Transfer)**.** *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k=(i,j)\in E$, $v_k = \|[S_c]_{:,k}\|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A\setminus k)\|_2$ the discrete damage from fixed-normalization removal of edge k. Assume all single-edge removals are subcritical: $\sqrt{2}\max_k w_k < \varepsilon_{\mathrm{crit}}$.*

(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k\cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2}\cdot w_k^2, \quad (9)$$

where $L_J \leq \|W\|_2^2\,\|z^*\|$ is the second-order curvature constant of $z^*(A)$, finite at the fixed point.

(b) Ranking preservation (pairwise-sufficient). *For two edges k_1,k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1}\geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1}-v_{k_2}}{L_J/(1-\kappa)^2}. \quad (10)$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1\neq k_2}(v_{k_1}-v_{k_2})\cdot(1-\kappa)^2/L_J$.

Proof in section A.4. Modelling deletion as fixed-normalization masking gives $\|\Delta z^*\| \approx w_k v_k$ from the S_c construction; the second-order remainder $|R_k|$ is bounded via the IFT curvature term, which carries two resolvents and the finite equilibrium norm, and (b) follows by comparing the first-order gap to that remainder.

Empirically $|R_k|$ is 2–10$\times$ smaller than $L_J w_k^2$, the constrained S_c tightens the bound from 0.31 to 1.00, and the edge-weighted score $w_k v_k$ of (a) reaches $\tau = +\mathbf{0.996}$ against brute-force N-1 on full-graph Amazon Photo (10 seeds, section 5.3), corroborating the first-order bridge at scale.

3.4 Generalization to Explicit GNNs

Proposition 4 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K\circ\cdots\circ g_1(X,A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial\mathrm{vec}(Z_K)}{\partial\mathrm{vec}(A)} = \sum_{l=1}^{K}\left(\prod_{k=l+1}^{K} J_z^{(k)}\right) J_A^{(l)}, \quad (11)$$

where $J_z^{(k)} = \partial g_k/\partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l/\partial\mathrm{vec}(A)$. Then:
(a) *The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K)\cdot\|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \leq \sum_{l=1}^{K}\left(\prod_{k=l+1}^{K}\left\|J_z^{(k)}\right\|_2\right)\left\|J_A^{(l)}\right\|_2. \quad (12)$$

(b) *The constrained construction S_c and per-node radius r_v (Proposition 2) apply identically with S_K in place of S.*

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2\,(1-\kappa^K)/(1-\kappa)$ converges to the IGNN bound as $K\to\infty$. Explicit GNNs thus inherit the full pipeline (empirical tightness 0.99–1.02 across the suite, section 5.3); only the closed-form $\varepsilon_{\mathrm{crit}}$ is restricted to the contractive subclass.

4 AEGISConstruction

The pipeline makes the theory of section 3 practical: it queries S_c matrix-free, never materialising $S\in\mathbb{R}^{Nd\times N^2}$ or forming the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta\hat{A}^\star$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\mathrm{crit}}$ and the regime characterisation of Theorem 1 for IGNN-class operators. A dense path ($N \leq 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to N=7,650) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 (section C) gives the matrix-free routing.

Matrix-free operator. The key to scaling is to never build S_c: we only apply it to vectors, needing $O(Nd)$ memory rather than the $O((Nd)^2)$ a dense S_c costs. Concretely, $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^{K} J_z^k b$ with depth K adaptive in κ (early stop at $\left\|J_z^k b\right\| < 10^{-6} \|b\|$), each Neumann term and $J_A \operatorname{vec}(\delta A)$ being forward-mode JVPs in $O(Nd)$ time and P_c applied implicitly, so neither S_c nor S is formed; the dense path ($N \leq 200$) materialises J_z, J_A and uses a direct solve. The rSVD (Halko, Martinsson, and Tropp 2011) (k=p=10, n_{iter}=2) returns (u_1, σ_1, v_1) for the direction $\delta\hat{A}^\star$=$\varepsilon \operatorname{sym}(P_c v_1)$ (Proposition 1), with $\operatorname{sym}(M)$=$(M+M^\top)/2$ keeping $\delta\hat{A}^\star$ symmetric (section 2); column norms give $\{v_{ij}\}$ and per-node radii follow Proposition 2. The well-separated singular spectrum (gap $(\sigma_1 - \sigma_2)/\sigma_1$=0.39–0.50; fig. 4) is why one rSVD query matches iterative attackers (section 5.2).

Cost. Each $S_c v$ is $O(K \cdot Nd)$ time and $O(Nd)$ memory ($K \in [20, 50]$ for κ<0.8); removing the $O((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section D.3).

5 Experiments

Our evaluation tests four claims in turn: the contractivity assumptions hold on trained models (section 5.1); the one-query SVD direction matches iterative attackers (section 5.2); the continuous S_c ranking transfers to discrete edge removal across architectures (section 5.3); and the same object audits a deployed fraud detector (section 6).

Setup. We evaluate AEGISon 6 datasets across 4 domains with 10 seeds each (PyTorch (Paszke et al. 2019)); $\pm$ are standard deviations over seeds. IGNN (Gu et al. 2020) uses $F(Z)$=$\operatorname{ReLU}(\hat{A}ZW^\top + U(X))$, $W \in \mathbb{R}^{64\times 64}$ spectral-normalised (Miyato et al. 2018), Adam (lr 0.01); the constraint costs $\sim$6% Cora accuracy. Datasets: Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), Amazon Fraud (Dou et al. 2020). Baselines: smoothing (Gaussian σ=0.01), Mettack, and GR-BCD/PR-BCD (Geisler et al. 2021). All seven architectures appear only in fig. 3 and table 5; the rest use IGNN. Full tables, ablations, and timing are in section D; reproducibility in section F.

5.1 Cross-Domain Adversarial Analysis

Two prerequisites must hold: the trained models sit in the contractive regime, and AEGIS's perturbations beat chance. Table 2 confirms both, verifying (A3) on every benchmark (κ<1) and reporting per-dataset $\varepsilon_{\text{crit}}$ alongside a 3.2–4.1× attack advantage over random. Unless stated otherwise, IGNN experiments use 50-node BFS ego-subgraphs (highest-degree centre) reporting *local vulnerability*, graph-level assessment using the matrix-free pipeline (section D.3); all bounds use κ=$\|J_z\|_2$.

The first-order shift bounds agree with actual equilibrium shifts within 1% at the small budgets that matter for early warning, loosening to at most 39% on citation graphs (within 7% on WikiCS and Amazon) even at ε=0.20. AEGISbeats degree-proportional, edge-betweenness, and spectral rankings at ε=0.01 (+6–148% AtkAdv) and inflicts 3–10× more equilibrium damage than Mettack (Zügner and Günnemann 2019a) in the early-warning regime (**149/150** paired wins, p<10^{-43}); GR-BCD/PR-BCD and discrete-removal curves are in section D.2.

Table 2: Per-dataset IGNN characterisation (10 seeds). κ=$\|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}}$=$(1-\kappa)/\|W\|_2$ is the local critical budget. AtkAdv: AEGIS/random damage. Cov%: subgraph nodes certified first-order-safe at ε=0.05 (r_v>0.05).

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	$77.5_{\pm 1.7}$	$.33_{\pm .14}$	$.66_{\pm .15}$	$3.6_{\pm .6}$	$83_{\pm 10}$
Citeseer	$66.0_{\pm 0.7}$	$.59_{\pm .02}$	$.41_{\pm .02}$	$4.1_{\pm .5}$	$94_{\pm 5}$
Pubmed	$78.9_{\pm 0.7}$	$.41_{\pm .11}$	$.59_{\pm .11}$	$3.2_{\pm .4}$	$76_{\pm 18}$
Amazon	$94.8_{\pm 0.3}$	$.14_{\pm .02}$	$.86_{\pm .02}$	$3.8_{\pm .2}$	$86_{\pm 8}$
WikiCS	$77.9_{\pm 0.4}$	$.34_{\pm .04}$	$.66_{\pm .04}$	$3.8_{\pm .4}$	$78_{\pm 4}$

5.2 Attack Efficiency and the One-Query Direction

AEGIS's attack is one-query and analytic, not a peak optimiser. The SVD direction maximises first-order equilibrium shift by construction (Proposition 1); what matters is non-circularity and cost. A transfer direction from a *separately trained* surrogate (zero shared gradients) recovers 99% of the one-query damage ($\cos$=0.99), versus 44±4% for a 512-query black-box search, so it is model-intrinsic, not a gradient artifact. Per query it delivers **74–156**× the equilibrium damage of a 50-step PGD (Madry et al. 2018); classification-loss PGD inflicts 15–70% less at 50× the cost. Table 3 (section D.1) gives the full four-quadrant comparison and contrasts r_v with randomized/localized smoothing radii; budget-heavy attackers (GR-BCD) exceed AEGISonly at the larger budgets they target (section D.2). Per-edge finite differences reproduce the S_c column-norm ranking (τ=0.999), and iterative recomputation closes ≤ 3 pp of the static-vs-greedy gap at $k\times$ compute, so the static ranking is the headline.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a reliable first-order screen (section 3.1). Figure 2 reports prediction-flip fractions vs. ε (95% CI bands): Citeseer and WikiCS stay below 2% throughout; Cora reaches 7.6% at ε=0.20; Pubmed is the right-skewed outlier (median 7.8%, mean 10.3% at ε=0.10; 27.4% at ε=0.20).

5.3 Continuous-to-Discrete Transfer Across Architectures

The construction is not tied to implicit models: for K-layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT[†] (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form S_K=$\partial\operatorname{vec}(Z_K)/\partial\operatorname{vec}(A)$

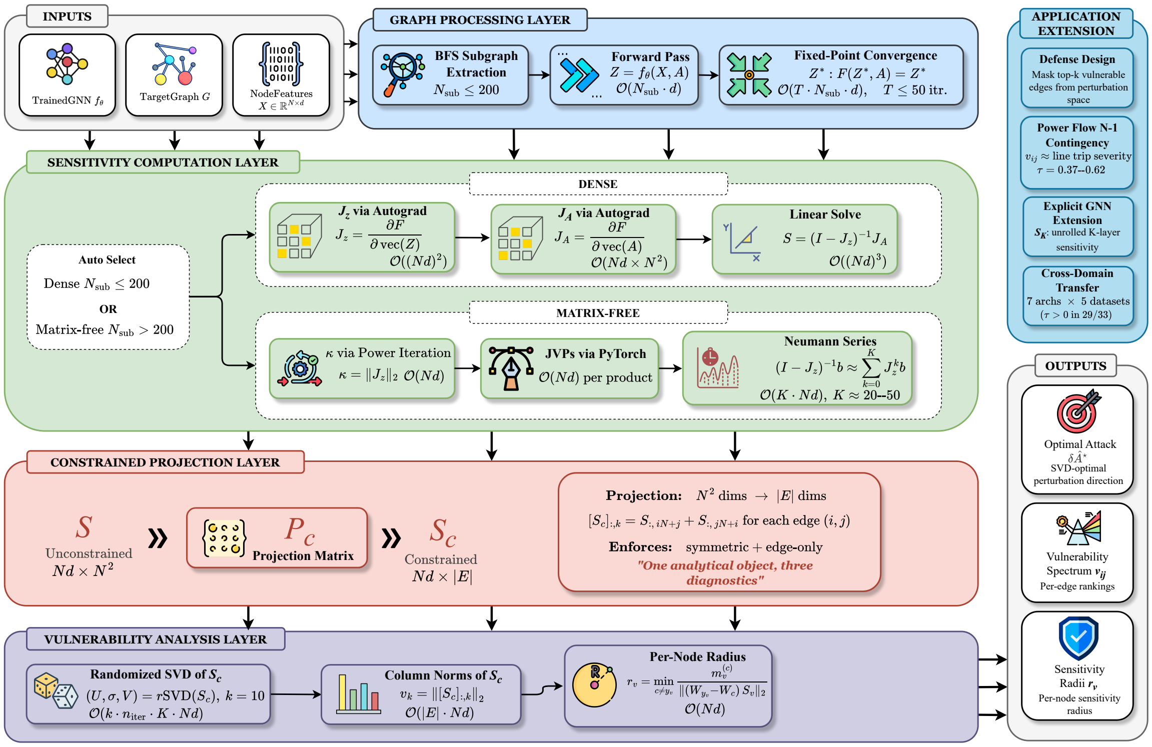


Figure 1: The four-stage AEGISpipeline (section 4): (1) graph processing (BFS subgraph or full-graph fixed-point pass); (2) sensitivity (dense Jacobian for $N \le 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \to |E|$ via P_c; (4) analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJPs without materialising it, scaling to N=7,650.

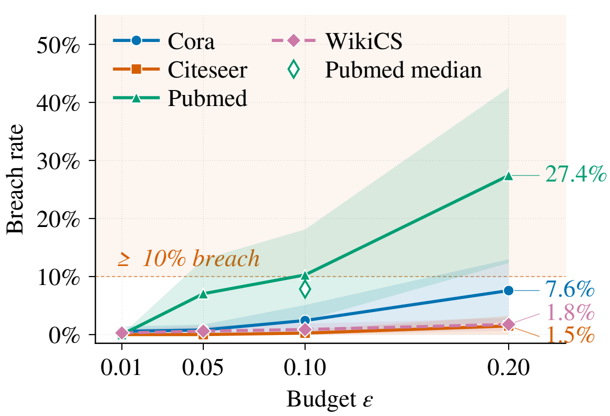


Figure 2: Breach rate (% of nodes flipped under S_c-optimal perturbation) vs. ε (10 seeds; mean, 95% CI bands).

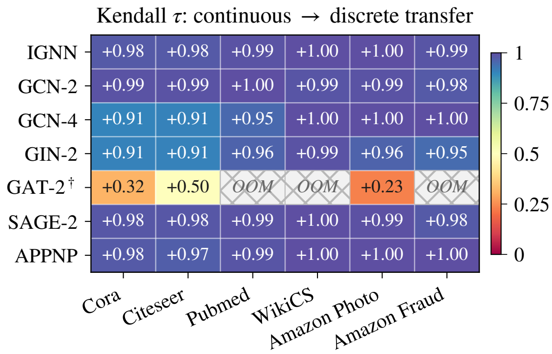
via finite differences and build S_c identically (Proposition 4); IGNN additionally carries Theorem 1, the explicit models getting the tool without the regime characterisation (per-architecture detail in table 5, section D.5).

The headline test is whether the continuous S_c ranking predicts discrete N-1 removal. Ranking edges by the edge-weighted $A_{ij}v_{ij}$ of Proposition 3, fig. 3 reports its Kendall τ over 6 datasets and 7 architectures (390 runs): *all* $\mathbf{39/39}$ *cells positive*, median τ=$+\mathbf{0.99}$ (p<10^{-5}), vs. 34/39 unweighted. Weighting matters most at full-graph scale: on Amazon Photo (N=7,650, κ≈1.00) it reaches τ= $+\mathbf{0.996}\pm\mathbf{0.002}$. The score adds rank information beyond the weight (beating a pure-weight baseline by $\Delta\tau$≈+0.25, up to +0.65 on the fraud graph), and Proposition 3's pairwise condition holds for only 47–62% of pairs, so the global τ is empirical. Per-architecture detail, GAT† scope, robust backbones, and the cross-dataset breakdown are in section D.5.

The phase-transition sweep (a 2–4× spectral-radius margin to criticality holding even as $\varepsilon_{\text{crit}}$ falls 230×) and the scalability study (matrix-free reaching N=7,650 at 365 s/5.5 GB, σ_1 within 0.03% at N=200) are in section D.3; subgraph-size, hyperparameter, and edge-protection ablations (vulnerability is *delocalized*, so risk is localised for triage rather than removed by masking a few edges) are in section D.4.

6 Case Study: Auditing a Fraud Detector

GNN fraud detectors are a prime adversarial target: a flagged account can evade detection by perturbing a few behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGISaudits such a de-



† *GAT-2: 2-layer attention; OOM on Pubmed, WikiCS, Amazon Fraud.*

Figure 3: Continuous-to-discrete transfer τ (10-seed means, 390 runs; per-cell sd 0.0003–0.041), ranking edges by the edge-weighted $A_{ij}v_{ij}$ of Proposition 3. All $39/39$ cells positive (median $+0.99$); cross-hatch: GPU OOM. GAT-2† is lower under weighting (it transfers better unweighted; see section D.5).

tector directly — no labels, no retraining — reading from *one* S_c query which edges most threaten its predictions. On an Amazon Fraud cluster the edge-weighted ranking $A_{ij}v_{ij}$ (Proposition 3) reproduces the brute-force single-edge-removal damage ranking (τ=1.0 on the cluster) at one query's cost rather than $|E|$ reconvergences. The audit is validated at scale and *non-circularly*: the ranking transfers across the full fraud graph (fig. 3), precisely where uniform edge weights are uninformative so v_{ij} adds the most rank information ($\Delta\tau$ up to $+\mathbf{0.65}$), and the SVD direction transfers from a separately trained surrogate ($\cos = 0.99$; section 5.2), so v_{ij} reflects model-intrinsic vulnerability rather than a gradient artifact. The detailed walk-through, with the audited cluster and its fault-line diagram, is in section E.

7 Related Work

Each of three threads supplies at most one of AEGIS's three diagnostics. **Structural attacks** — Nettack (Zügner, Akbarnejad, and Günnemann 2018), Mettack (Zügner and Günnemann 2019a), GR-BCD/PR-BCD (Geisler et al. 2021) and variants (Wu et al. 2019; Xu et al. 2019a) — optimise surrogate gradients to flip predictions but give no analytic maximally-sensitive direction and no per-node radius; sober-look studies (Mujkanovic et al. 2022; Gosch et al. 2023) flag evaluation pitfalls, whose adaptive-attack discipline we follow in section D.4.

Certified robustness and defense. Convex relaxations (Zügner and Günnemann 2019b), smoothing (Bojchevski, Klicpera, and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023), collective (Schuchardt et al. 2021) and majority-vote (Li and Wang 2025) certificates, and weight-norm bounds (Abbahaddou et al. 2024) certify per-node robustness but do not identify vulnerability-driving edges; empirical defenses (Zhu et al. 2019; Jin et al. 2020; Zhang and Zitnik 2020) harden models without surfacing per-edge sensitivity. Graph conformal prediction (Zargarbashi, Antonelli, and Bojchevski 2023) certifies a *fixed* graph; AEGIS-Conformal instead covers the adversarial ε-ball via the S_c score-shift bound (section 3.1), complementary to discrete-edit certifiers (section D.2).

Implicit networks and influence functions. DEQs (Bai, Kolter, and Koltun 2019), IGNN (Gu et al. 2020), monotone equilibria (Winston and Kolter 2020), and well-posedness analyses (El Ghaoui et al. 2021) address *input* sensitivity $\partial z^*/\partial x$; AEGIStargets *structural* sensitivity $\partial Z/\partial A$. It descends from the Koh–Liang influence-function lineage (Koh and Liang 2017) and implicit-differentiation hyperparameter optimisation (Lorraine, Vicol, and Duvenaud 2020; Gould, Hartley, and Campbell 2021), but those apply IFT to loss perturbations whereas AEGISapplies it to a *structural-parameter* perturbation via P_c. GNN explainers (Ying et al. 2019) give per-edge importance for *fidelity*, not vulnerability; matrix perturbation theory (Stewart and Sun 1990; Trefethen and Embree 2005) supplies the pseudospectral toolkit; and curvature/over-squashing (Topping et al. 2022; Di Giovanni et al. 2023) and stability bounds (Gama, Bruna, and Ribeiro 2020; Kenlay, Thanou, and Dong 2021) tie edge structure to sensitivity only in aggregate, whereas AEGISlocalises it per edge and node.

8 Conclusion

From one analytical object, the constrained sensitivity matrix S_c, AEGIS*audits* a GNN (per-edge rankings, the SVD-optimal direction, per-node radii), *certifies* it (AEGIS-Conformal, section 3.1), and *defends* it (regularizing $\sigma_1(S_c)$). The construction is matrix-free and applies to any GNN with continuous edge-weight-modulated message passing; for contractive implicit models it adds a regime characterisation (Theorem 1) that holds as a conservative safety boundary.

Limitations. The per-node radii r_v are first-order thresholds; AEGIS-Conformal (section 3.1) currently runs densely at N=200 (a matrix-free conformal pass is future work). Standard GAT and binary-mask architectures fall outside the framework, and full-graph matrix-free analysis is the recommended graph-level path (section 5.3); opting into the formal $\varepsilon_{\text{crit}}$ track accepts a $\sim$6% accuracy cost, on par with certified-robustness trade-offs. Insertion attacks (Nettack-class) are scoped out, extending once the edge basis is enlarged to $\bar{E} \supseteq E$ (section 2). Finally, AEGISaudits a model's vulnerability, not ground-truth physics: a contractive surrogate always converges to a fixed point, so it cannot represent voltage-collapse contingencies and complements rather than replaces domain analyses such as power-flow contingency screening (Donon et al. 2019; Nakiganda et al. 2023; Varbella, Gjorgiev, and Sansavini 2024).

Disclosure protocol. We propose a 90-day coordinated-notification window before release, the attack-direction reconstruction (algorithm 1, steps 3–4) gated behind institutional-ethics review, and the diagnostic-only path (r_v, v_{ij} scores) released unconditionally: a score *ranks* sensitive edges but is not itself an attack, the damaging δA^* (Proposition 1) coming only from the gated reconstruction.

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A Proofs for the Adversarial Equilibrium Theory

This appendix collects the proof bodies for the results stated in section 3. We restate each result in brief for self-containment.

A.1 Proof of Theorem 1

Recall Theorem 1 characterises three regimes of the perturbation budget ε for an IGNN-class operator under (A1)–(A3), with critical budget $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\left\|W\right\|_2$.

Proof. **(a).** Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\left\|\delta \hat{A}\right\|$ (the boundary set

$\{\hat{A} : z^*(\hat{A})$ lies on a region face$\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels (Bolte and Pauwels 2021) yields a conservative IFT on each region. With $J_z = \operatorname{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \left\|\hat{A}\right\|_2 \|W\|_2$, applying IFT to $G(z,\hat{A}) = z - F(z,\hat{A}) = 0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \operatorname{vec}(\delta\hat{A}) + O(\left\|\delta\hat{A}\right\|^2), \tag{13}$$

with $J_A = \partial F / \partial \operatorname{vec}(A)$ and Neumann $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ yielding (5). For contractivity preservation, $\|J_z'\|_2 \leq (\left\|\hat{A}\right\|_2 + \varepsilon)\|W\|_2$, so $\varepsilon < \varepsilon_{\mathrm{crit}}$ guarantees $\|J_z'\|_2 < 1$.

(b)–(c). For any M, $(I-M)^{-1}$ has eigenvalues $(1-\lambda_i(M))^{-1}$, hence $\left\|(I-M)^{-1}\right\|_2 \geq \rho((I-M)^{-1}) = 1/\min_i |1-\lambda_i(M)|$ *unconditionally*. Along the worst-case direction $\|J_z'\|_2 \leq (\left\|\hat{A}\right\|_2 + \varepsilon)\|W\|_2$, so $1 - \|J_z'\|_2 \geq \|W\|_2 (\varepsilon_{\mathrm{crit}} - \varepsilon)$ and, since $\rho \leq \|\cdot\|_2$, $\varepsilon < \varepsilon_{\mathrm{crit}}$ certifies $\|J_z'\|_2 < 1$. The certificate is one-sided: $\|J_z'\|_2 \to 1$ need not drive $\min_i |1-\lambda_i| \to 0$ (e.g. $M = \operatorname{diag}(-s, 0)$ has $\|M\|_2 = s$ yet $\left\|(I-M)^{-1}\right\|_2 = 1$). When J_z' is normal with dominant real-positive eigenvalue $\lambda^\star = \|J_z'\|_2$ (the Perron mode for symmetric $\hat{A}, W$), $\min_i |1-\lambda_i| = 1 - \lambda^\star = \|W\|_2 (\varepsilon_{\mathrm{crit}} - \varepsilon)$ yields the $\Omega(1/(\varepsilon_{\mathrm{crit}} - \varepsilon))$ rate; a negative or complex dominant eigenvalue keeps $\min_i |1-\lambda_i|$ away from 0 (the same $\operatorname{diag}(-s,0)$), so $\varepsilon_{\mathrm{crit}}$ only lower-bounds the threshold, slack controlled by η (Observation 1). For $\varepsilon \geq \varepsilon_{\mathrm{crit}}$ Banach contraction fails and the iteration may converge elsewhere, oscillate, or diverge. □

A.2 Proof of Observation 1

Proof. $J_z = \hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ (Stewart and Sun 1990); symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}}) = 1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\left\|(I-M)^{-1}\right\|_2 \leq \kappa(V)/(1-\rho(M))$ yields $\eta \leq \kappa(V_W)$. □

A.3 Proof of Proposition 2

Proof. By Theorem 1(a), $\Delta z_v^* \approx S_v \operatorname{vec}(\delta A)$, so for each competitor $|\Delta(f_{y_v} - f_c)| \leq \|(W_{y_v} - W_c)S_v\|_2 \|\delta A\|_F$ by Cauchy–Schwarz; the prediction survives while every margin stays positive, i.e. $\|\delta A\|_F < m_v^{(c)} / \|(W_{y_v} - W_c)S_v\|_2$ for all c, whose minimum is r_v. Intuitively, r_v is the smallest margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structure); high-margin, low-sensitivity nodes get large radii. □

A.4 Proof of Proposition 3

Proof. **(a).** We model deletion as *fixed-normalization* masking, $[\delta\hat{A}]_{ij} = [\delta\hat{A}]_{ji} = -w_k$ with the degree matrix D held fixed, matching the continuous edge-weight perturbation, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. (recompute-normalization adds an $O(d_i^{-1})$ incident-edge rescaling, negligible off low-degree nodes; agreement in section 5.1.) For the remainder, the IFT gives $\partial z^* / \partial \operatorname{vec}(A) = (I - J_z)^{-1} J_A$, and the dominant curvature term $(I-J_z)^{-1}(\partial J_z / \partial \operatorname{vec}(A))(I-J_z)^{-1} J_A$ carries two resolvents $(1-\kappa)^{-1}$, two factors $\|W\|_2$, and the equilibrium norm $\|z^*\|$ (since $\partial J_z / \partial \operatorname{vec}(A) \propto W$ and $J_A \propto z^* W^\top$), so $L_J \leq \|W\|_2^2 \|z^*\|$, finite since $\|z^*\| \leq \|X_{\mathrm{proj}}\|/(1-\kappa)$ (the κ-contraction of F at the fixed point, A3), giving $\left\|\partial^2 z^* / \partial \operatorname{vec}(A)^2\right\|_2 \leq (1-\kappa)^{-2} L_J$. The path meets finitely many ReLU regions (simultaneous crossings non-generic, measure zero); summing the second-order remainder over these regions, with the conservative IFT (Bolte and Pauwels 2021) on each, gives $|R_k| \leq L_J w_k^2 / (2(1-\kappa)^2)$. **(b).** From (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J / (2(1-\kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \geq w_{k_2}$, the first term is positive and (10) ensures the first-order gap dominates. □

B Constant-Factor Two-Sided Boundary Theorem

This appendix gives the full statement and proof of Theorem 2 (the constant-factor two-sided boundary cited in section 3), together with the empirical nonlinear extension (Observation O1). The lower side (i) is the sound certificate sold by the paper; the upper side (ii)–(iii) is proved for the all-active operator (A4).

Notation and standing assumptions. $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$ is the symmetric normalized adjacency; W the tied weight; ϕ the elementwise activation; z^* the equilibrium; the all-active Jacobian is $J_z = \operatorname{diag}(\phi')(\hat{A} \otimes W)$, $\kappa := \|J_z\|_2$, $\hat{A}' = \hat{A} + \delta\hat{A}$, $J_z' = \operatorname{diag}(\phi')(\hat{A}' \otimes W)$.

Definition 1 (Operator and certified regime)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A} Z W^\top + X_{\mathrm{proj}})$ with:*

- *(A1) ϕ is 1-Lipschitz (ReLU or any such), so $\|\operatorname{diag}(\phi')\|_2 \leq 1$ everywhere; nonsmoothness at 0 is handled by the conservative IFT (Bolte and Pauwels 2021).*
- *(A2) $\|W\|_2 \leq c$ via spectral normalization (controls the* numerator *of g_W only; see the a-posteriori remark below).*
- *(A3) The trained model is contractive: $\kappa = \|J_z\|_2 < 1$, reported and audited post-training ($\kappa \in [0.14, 0.59]$ across our suite). Equivalently $\left\|\hat{A}\right\|_2 \|W\|_2 < 1$.*
- *(A4, all-active) On the operating set $\phi' \equiv 1$, so $J_z = \hat{A} \otimes W$. The* upper *bound (ii)–(iii) of Theorem 2 is proved* for this operator*; the lower bound (i) does* not *use (A4).*

Audited (a-posteriori) gap quantities. Define $g_W := \|W\|_2 / \rho(W) \geq 1$ and $g_A := \left\|\hat{A}\right\|_2 / \rho(\hat{A}) = 1$ ($\hat{A}$ symmetric). g_W is the *nonnormality* of W, stated as an audited per-model constant, not a quantity controlled a priori by training: spectral normalization (A2) bounds $\|W\|_2$ but not $\rho(W)$ away from 0, so in principle $\rho(W) \to 0$ at fixed $\|W\|_2$ sends $g_W \to \infty$. We *measure* g_W on each trained model ($g_W \in [1.19, 2.47]$ in our suite, with the a-posteriori certificate $g_W \leq \kappa_2(V_W)$, the eigenvector conditioning of W) and report C as a function of that value. Every constant in Theorem 2 is computable post-training from $\rho(W)$, $\|W\|_2$, $\left\|\hat{A}\right\|_2$ and the edge set.

Two budgets and the boundary.

$$\varepsilon_{\rm crit} := \frac{1-\kappa}{\|W\|_2} = \frac{1}{\|W\|_2} - \left\|\hat{A}\right\|_2 \quad \text{(norm certificate)},$$
$$\varepsilon_{\rm spec} := \frac{1}{\rho(W)} - \rho(\hat{A}) \quad \text{(all-active spectral break)}.$$

$\varepsilon_{\rm br}^{\rm all}$:= the infimal $\left\|\delta\hat{A}\right\|_F$ over feasible (symmetric, edge-supported) perturbations at which the *all-active* operator loses contraction, $\rho(\hat{A}' \otimes W) \geq 1$; the superscript marks that the upper-bounded object is the all-active linearization, not the deployed ReLU model's stability boundary.

Edge-support alignment. Let P_E be the orthogonal projection onto the symmetric, edge-supported subspace $\mathcal{S}_E := \{M = M^\top : M_{ij} = 0 \text{ if } (i,j) \notin E \cup \{(i,i)\}\}$. With u_1 a unit top eigenvector of $\hat{A}$, define $g_E := P_E(u_1 u_1^\top)$, $\sigma_E := \|g_E\|_F \in [0,1]$, and (when $\sigma_E > 0$) the unit feasible direction $B := g_E/\sigma_E$. Then $\beta := \langle u_1, Bu_1\rangle = \sigma_E^2/\sigma_E = \sigma_E \in (0,1]$, using $\langle u_1, P_E(u_1u_1^\top)u_1\rangle = \|g_E\|_F^2 = \sigma_E^2$ (P_E an orthogonal projection, $u_1u_1^\top$ symmetric). Thus the alignment β equals the edge-supported Frobenius mass $\sigma_E = \left\|P_E(u_1u_1^\top)\right\|_F$ of the rank-one critical direction, strictly positive whenever the top Perron mode places mass on the edge set (always true for a connected graph, $u_1 > 0$ entrywise, $E \neq \emptyset$), making the hypothesis $\beta > 0$ an observable rather than an assumption.

Theorem 3 (Constant-factor two-sided bracket for the all-active IGNN contraction boundary, full statement)**.** *Let Definition 1 hold with $\varepsilon_{\rm crit} > 0$ (certified regime) and $\hat{A}$ symmetric, and let $\beta = \sigma_E \in (0,1]$ be the edge-supported mass of the critical mode. Then:*

(i) ***Lower side — sound certificate (any 1-Lipschitz ϕ; no (A4)).*** *For every feasible $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F \leq \varepsilon < \varepsilon_{\rm crit}$, the perturbed operator $F_\theta(\cdot, \hat{A}+\delta\hat{A})$ is an $\|\cdot\|_2$-contraction ($\|J_z'\|_2 < 1$) with a unique equilibrium and $\rho(J_z') < 1$. Hence $\varepsilon_{\rm br} \geq \varepsilon_{\rm crit}$; in particular $\varepsilon_{\rm br}^{\rm all} \geq \varepsilon_{\rm crit}$.*

(ii) ***Upper side — all-active matching attack (requires (A4)).*** *There is a feasible rank-one $\delta\hat{A}^\star = (\varepsilon_{\rm spec}/\beta)\, B \in \mathcal{S}_E$ with $\left\|\delta\hat{A}^\star\right\|_F = \varepsilon_{\rm spec}/\beta$ that drives the* all-active *spectral radius to 1; hence $\varepsilon_{\rm br}^{\rm all} \leq \varepsilon_{\rm spec}/\beta$. In the unconstrained-symmetric threat model ($P_E = I$, $\beta = 1$) the construction is* exact*: $\varepsilon_{\rm br}^{\rm all} = \varepsilon_{\rm spec}$ with extremizer $\varepsilon_{\rm spec}\, u_1u_1^\top$.*

(iii) ***Constant-factor enclosure.*** *$\varepsilon_{\rm spec} \leq C\,\varepsilon_{\rm crit}$ with the explicit, dimension-free, a-posteriori-computable constant*

$$C := g_W\,\frac{1+\kappa}{1-\kappa} = \frac{\|W\|_2}{\rho(W)} \cdot \frac{1 + \left\|\hat{A}\right\|_2 \|W\|_2}{1 - \left\|\hat{A}\right\|_2 \|W\|_2}.$$

Combining with (i)–(ii), the all-active contraction boundary obeys

$$\boxed{\varepsilon_{\rm crit} \;\leq\; \varepsilon_{\rm br}^{\rm all} \;\leq\; \frac{C}{\beta}\,\varepsilon_{\rm crit}}. \tag{14}$$

The bracket is non-vacuous whenever $\beta = \sigma_E > 0$ (every connected graph); on the suite $\beta \approx 0.62$, giving $C/\beta \lesssim 16$. The β-free form $\varepsilon_{\rm crit} \leq \varepsilon_{\rm br}^{\rm all} \leq C\,\varepsilon_{\rm crit}$ holds exactly for the unconstrained-symmetric model ($\beta = 1$); we report C/β as the certified edge-feasible constant and C as the $\beta = 1$ specialization.

(iv) ***Exact boundary (rate-sharp regime), separated equality conditions.*** *(a)* Endpoints coincide: $\varepsilon_{\rm crit} = \varepsilon_{\rm spec} \iff g_W = 1$ *(W normal); κ, β irrelevant. (b)* All three coincide: $\varepsilon_{\rm crit} = \varepsilon_{\rm br}^{\rm all} = \varepsilon_{\rm spec} \iff g_W = 1$ *and $\beta = 1$ (the exact extremizer $\varepsilon_{\rm spec}\, u_1u_1^\top$ is edge-feasible). (c)* Slack constant equals one: $C/\beta = 1 \iff g_W = 1,\ \beta = 1,$ *and $\kappa \to 0^+$ (kills the $\frac{1+\kappa}{1-\kappa}$ inflation).*

This is a constant-factor (dimension-free, $O(1)$) two-sided characterisation, rate-sharp in the normal/edge-aligned regime, with enclosure constant C/β reaching $\sim$10–16$\times$ on our suite (random-instance $\varepsilon_{\rm spec}/\varepsilon_{\rm crit} \in [1.02, 18.2]$, median 2.25); it is not "tight" in the matching-bound sense except in the corner (iv)(b). Two structural identities underlie it (both numerically exact over 4,000–6,000 random certified-regime instances): the *all-active spectral law* $\rho(J_z') = \rho(\hat{A}')\rho(W)$, $\|J_z'\|_2 = \left\|\hat{A}'\right\|_2 \|W\|_2$ (A4 only); and the *additive gap identity* $\varepsilon_{\rm spec} - \varepsilon_{\rm crit} = (\frac{1}{\rho(W)} - \frac{1}{\|W\|_2}) + (\left\|\hat{A}\right\|_2 - \rho(\hat{A})) \geq 0$, the slack supplied entirely by W's non-normality when $\hat{A}$ is symmetric (verified to 2.7×10^{-15}).

Proof. **Setup.** Feasible perturbations lie in $\mathcal{S}_E$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$. We use Weyl, $\rho(\hat{A}') \leq \rho(\hat{A}) + \left\|\delta\hat{A}\right\|_2$, $\left\|\hat{A}'\right\|_2 \leq \left\|\hat{A}\right\|_2 + \left\|\delta\hat{A}\right\|_2$, and $\left\|\delta\hat{A}\right\|_2 \leq \left\|\delta\hat{A}\right\|_F = \varepsilon$. Since $\hat{A}$ and the extremal $\delta\hat{A}$ are symmetric, $\rho(\hat{A}') = \left\|\hat{A}'\right\|_2$ and Weyl holds with equality along an aligned rank-one mode.

(i) Lower side, fully nonlinear. By (A1) $\|\mathrm{diag}(\phi')\|_2 \leq 1$; the spectral norm is sub-multiplicative and Kronecker-multiplicative, so $\|J_z'\|_2 \leq \|\mathrm{diag}(\phi')\|_2 \left\|\hat{A}' \otimes W\right\|_2 \leq \left\|\hat{A}'\right\|_2 \|W\|_2 \leq (\left\|\hat{A}\right\|_2 + \varepsilon)\|W\|_2$. For $\varepsilon < \varepsilon_{\rm crit} = \frac{1}{\|W\|_2} - \left\|\hat{A}\right\|_2$ this is < 1. A spectral-norm contraction has a unique fixed point (Banach) and $\rho(J_z') \leq \|J_z'\|_2 < 1$, so no feasible $\varepsilon < \varepsilon_{\rm crit}$ breaks contraction of the deployed model: $\varepsilon_{\rm br} \geq \varepsilon_{\rm crit}$. No (A4) used.

Additive gap. $\varepsilon_{\rm spec} - \varepsilon_{\rm crit} = (\frac{1}{\rho(W)} - \frac{1}{\|W\|_2}) + (\left\|\hat{A}\right\|_2 - \rho(\hat{A})) \geq 0$ using $\rho \leq \|\cdot\|_2$; with $\hat{A}$ symmetric the second bracket is 0, so $\varepsilon_{\rm crit} \leq \varepsilon_{\rm spec}$.

(ii) Upper side, all-active matching attack (A4). Let u_1 be a unit top eigenvector of symmetric $\hat{A}$. With the rank-one symmetric direction $t\,u_1u_1^\top$ ($\left\|u_1u_1^\top\right\|_F = 1$),

$\rho(\hat{A} + t\,u_1 u_1^\top) = \rho(\hat{A}) + t$ exactly. Under (A4), $\rho(J_z') = (\rho(\hat{A}) + t)\rho(W)$ reaches 1 at $t = \varepsilon_{\text{spec}} = \frac{1}{\rho(W)} - \rho(\hat{A})$, so the unconstrained all-active break budget is exactly $\varepsilon_{\text{spec}}$ with extremizer $\varepsilon_{\text{spec}}\, u_1 u_1^\top$. For edge feasibility, project to $\mathcal{S}_E$ via $B = P_E(u_1 u_1^\top)/\sigma_E$, so $\beta = \sigma_E = u_1^\top B u_1 > 0$. The map $t \mapsto \lambda_{\max}(\hat{A} + tB) = \max_{\|v\|=1} v^\top(\hat{A} + tB)v$ is a pointwise maximum of affine functions, hence *convex*, so it lies above its tangent $\rho(\hat{A}) + \beta t$ at $t = 0$: $\rho(\hat{A} + tB) \geq \rho(\hat{A}) + \beta t$ for all $t \geq 0$ (numerically: $0/42000$ violations of this lower bound, vs. $42000/42000$ for a concave version). Choosing $\delta\hat{A}^\star = (\varepsilon_{\text{spec}}/\beta)B \in \mathcal{S}_E$ drives $\rho(\hat{A}') \geq \rho(\hat{A}) + \varepsilon_{\text{spec}} = \frac{1}{\rho(W)}$, so $\rho(J_z') \geq 1$ and $\varepsilon_{\text{br}}^{\text{all}} \leq \left\|\delta\hat{A}^\star\right\|_F = \varepsilon_{\text{spec}}/\beta$; exact when $\beta = 1$.

(iii) Closing the bracket. Dropping $-\rho(\hat{A}) \leq 0$ and using $\rho(\hat{A}) = \left\|\hat{A}\right\|_2$, $\varepsilon_{\text{spec}} \leq \frac{1}{\rho(W)} = g_W \frac{1}{\|W\|_2} = g_W(\varepsilon_{\text{crit}} + \left\|\hat{A}\right\|_2)$. Since $\left\|\hat{A}\right\|_2 = \kappa/\|W\|_2 = \kappa(\varepsilon_{\text{crit}} + \left\|\hat{A}\right\|_2)$ gives $\left\|\hat{A}\right\|_2 = \frac{\kappa}{1-\kappa}\varepsilon_{\text{crit}}$, we get $\varepsilon_{\text{spec}} \leq g_W \frac{1}{1-\kappa}\varepsilon_{\text{crit}} \leq g_W \frac{1+\kappa}{1-\kappa}\varepsilon_{\text{crit}} = C\,\varepsilon_{\text{crit}}$ (verified, zero violations over 6,000 instances). With (i) and (ii) the boxed bracket (14) follows.

(iv) Equality conditions. (a) The gap $(\frac{1}{\rho(W)} - \frac{1}{\|W\|_2}) \geq 0$ is zero iff $g_W = 1$; κ, β do not appear. (b) Add edge-feasibility of $\varepsilon_{\text{spec}}\, u_1 u_1^\top$, i.e. $\beta = 1$; then $\varepsilon_{\text{crit}} = \varepsilon_{\text{br}}^{\text{all}} = \varepsilon_{\text{spec}}$. (c) Additionally $\frac{1+\kappa}{1-\kappa} = 1$ requires $\kappa \to 0^+$. □

This recovers and sharpens Theorem 1: the old result is the lower side (i); (ii) supplies the matching all-active upper side; (iv) separates the three equality conditions.

Remark 2 (Observation O1: nonlinear break budget and the active fraction; empirical / conjecture, not part of Theorem 2)**.** *The deployed model is ReLU, not all-active; its measured nonlinear break budget $\varepsilon_{\text{reach}}$ exceeds the all-active $\varepsilon_{\text{spec}}$. Measured (seed-42 reachability, 50-node ego subgraph; $\left\|\hat{A}\right\|_F = 1.77$): at κ_0=0.5, $(\varepsilon_{\text{crit}}, \varepsilon_{\text{spec}}, \varepsilon_{\text{reach}}) = (1.000, 1.518, 2.273)$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}}$=1.50, $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}}$=2.27, resolvent γ=1.02; at κ_0=0.9, $(\varepsilon_{\text{crit}}, \varepsilon_{\text{spec}}, \varepsilon_{\text{reach}}) = (0.111, 0.629, 1.060)$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}}$=1.69, $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}}$=9.54, γ=1.02. Empirically $\varepsilon_{\text{reach}} \approx (1/a)\varepsilon_{\text{spec}}$ with a the ReLU active fraction ($a := \frac{1}{Nd}\mathbb{E}\,\|\text{diag}(\phi')\|_0 \in (0,1]$); with $a \approx 0.6$, $1/a \approx 1.5$–1.7, matching $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}} \in [1.5, 1.7]$. Composing with the bracket gives the observed envelope $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}} \lesssim C/a$, spanning roughly $[2.3, 10]$ and matching the measured $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}} \in [2.3, 9.5]$.*

This is a conjecture; two independent proof gaps remain open. *(1)* Masked-operator spectral scaling (unproved): *$\varepsilon_{\text{reach}} \approx \varepsilon_{\text{spec}}/a$ assumes the dominant eigenvalue of the masked Jacobian $D_a(\hat{A}' \otimes W)$ scales as $\approx a\,\rho(\hat{A}')\rho(W)$, but no such lemma holds in general (the active set correlates with the equilibrium and with u_1); this is a mean-field heuristic calibrated to two operating points. (2)* Linear-to-nonlinear bifurcation (empirical only): *the jump from "a linearized-Jacobian eigenvalue reaches 1" to "the true nonlinear fixed point destabilizes" is supported here only by the seed-42 run (a clean simple-pole divergence, γ=1.02). We present $\varepsilon_{\text{reach}} \approx (1/a)\varepsilon_{\text{spec}}$ as an empirical regularity organized by the same two knobs (g_W, a), not a proven extension.* Practical caveat: *$\varepsilon_{\text{reach}}$ is 60–128% of $\left\|\hat{A}\right\|_F$, i.e. 5–11× the largest realistic budget ($\varepsilon \leq 0.2$), so realistic-budget robustness is governed by the first-order $\sigma_1(S_c)$, not by proximity to criticality.*

Algorithm 1: AEGISmatrix-free vulnerability analysis. S_c is queried but never formed; the dense fallback ($N \leq 200$) builds S_c and runs an exact SVD.

Require: F, fixed point z^*, $\hat{A}$, edge set E, budget ε, rank k, Neumann tolerance tol; (opt.) head W, margins $m_v^{(c)}$.
Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^\star$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\text{crit}}$.
$\kappa \leftarrow \|J_z\|_2$; $\quad K \leftarrow \lceil \log(1/\text{tol})/\log(1/\kappa) \rceil$
$S_c\,v := \left(\sum_{j=0}^{K} J_z^j\right) J_A\, P_c\, v$ {matrix-free}
$(\sigma_1, v_1) \leftarrow$ rSVD(S_c, k) (Halko, Martinsson, and Tropp 2011)
$\delta\hat{A}^\star \leftarrow \varepsilon\,\text{sym}(P_c v_1)\,/\,\|\text{sym}(P_c v_1)\|_2$
$v_{ij} \leftarrow \|S_c\, e_{ij}\|_2$ for each $(i,j) \in E$
$r_v \leftarrow \min_{c \neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c)\, S_{c,v}\|_2$ {head only}
$\varepsilon_{\text{crit}} \leftarrow (1-\kappa)/\|W\|_2$ {Theorem 1}
return $(\sigma_1,\ \delta\hat{A}^\star,\ \{v_{ij}\},\ \{r_v\},\ \varepsilon_{\text{crit}})$

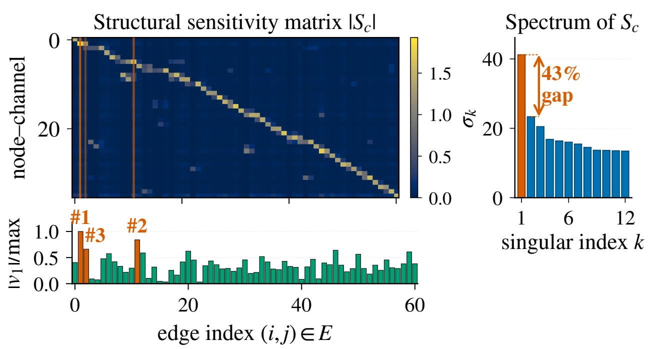

Figure 4: $|S_c|$ on a 50-node Cora ego-graph (IGNN, N=50, d=16, $|E|$=61, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, σ_1=41.2, 43% gap to σ_2 (cross-benchmark $(\sigma_1-\sigma_2)/\sigma_1 \in [0.39, 0.50]$; section 5.2). The well-separated leading mode is why one rSVD query matches iterative attackers.

C Matrix-Free Algorithm

Algorithm 1 summarises the matrix-free routing of the AEGISconstruction (section 4). S_c is queried via JVP/VJP but never materialised; the dense fallback ($N \leq 200$) builds S_c and runs an exact SVD.

Table 3: Equilibrium damage at ε=0.10 across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70_{\pm.79}$	$2.51_{\pm.48}$	$3.05_{\pm.70}$	$0.97_{\pm.25}$
Citeseer	$4.63_{\pm.71}$	$2.97_{\pm.42}$	$3.35_{\pm.57}$	$1.01_{\pm.15}$
WikiCS	$2.10_{\pm.22}$	$0.67_{\pm.11}$	$1.93_{\pm.20}$	$0.53_{\pm.09}$

D Detailed Experimental Results

This appendix gives the detailed tables and per-dataset breakdowns whose headline numbers appear in section 5.

D.1 Four-Quadrant Attack Comparison

Table 3 evaluates AEGISacross the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). The SVD direction maximises first-order equilibrium shift by construction (Proposition 1); what matters is non-circularity and cost. A transfer direction from a separately trained surrogate (zero shared gradients) recovers 99% of the one-query damage ($\cos = 0.99$), versus $44 \pm 4\%$ for a 512-query black-box search: model-intrinsic, not a gradient artifact. Per query it delivers 74–156$\times$ the damage of a 50-step PGD (Madry et al. 2018); classification-loss PGD inflicts 15–70% less at 50$\times$ the cost, IFT-gradient PGD (solver validation, not independent) 72–92%, and prediction flips stay 0–1.8% for all methods.

Radius vs. smoothing. Randomized smoothing (Bojchevski, Klicpera, and Günnemann 2020) gives constant-per-node base radii (0.123 vs. AEGIS's 0.078 at σ=0.05), and localized smoothing (Schuchardt et al. 2023) needs 10^3–10^4 MC/node; both lack edge structure. AEGISradii are instead structurally informative ($r_v \approx 0.10$ in dense regions, 0.01 at boundaries), a complementary point on the certificate-versus-diagnostic frontier.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20_{\pm 0.05}$ vs. AEGIS's $+0.32_{\pm 0.04}$) and stay near zero elsewhere; AEGISachieves 2–6$\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces. Per-edge finite differences reproduce the S_c column-norm ranking (τ=0.999); iterative S_c recomputation closes $\leq$ 3 pp of the static-vs-greedy gap at $k\times$ compute, so the static ranking is the headline.

D.2 Comparison with Prior Structural Baselines

Table 4 reports head-to-head numbers against the GR-BCD and PR-BCD structural attackers (Geisler et al. 2021). AEGISis a label-free one-query proxy; the question is how much of each gold-standard signal it retains. GR-BCD aligns with S_c on Pubmed (τ=+0.69) but decorrelates on Cora (τ=+0.16); the budgets shown ($k \leq 10$) are AEGIS's early-warning regime, where its one-pass diagnostic is competitive, while GR-BCD/PR-BCD dominate raw damage at the high budgets they target. AGNNCert (Li and Wang 2025) is a sound certificate against a bounded

Table 4: AEGISvs. external baselines (10 seeds). GR-BCD (Geisler et al. 2021) is the BCD-family scalable structural attacker (PR-BCD is the same paper's larger-budget variant, see text).

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD	Pubmed k=10	0.356	0.350	$+\mathbf{0.69}$
GR-BCD	Cora k=5	0.643	1.207	+0.16

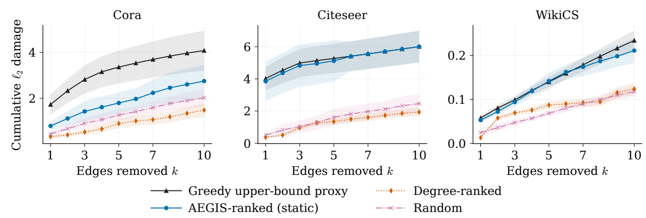

Figure 5: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS*matches* the ground-truth-aware Greedy attacker on Citeseer/WikiCS and recovers **54–67%** of Greedy's damage on Cora at k=5–10 *without any access to discrete ground truth*; degree-ranked stays below random on Cora at every k.

number of discrete edge edits (a majority vote over hash-partitioned sub-graphs); AEGISr_v instead bounds a *continuous* edge-weight budget $\left\|\delta \hat{A}\right\|_F$. The two target different threat models and are complementary rather than comparable on one radius axis: r_v is a sound first-order screen (no node breaches below it, section 5.2) that flags which edges a discrete certifier should prioritise. AEGISalso beats degree-proportional, edge-betweenness, and spectral rankings at ε=0.01 (+6–148% AtkAdv, Wilcoxon $p<0.001$) and inflicts 3–10$\times$ more equilibrium damage than Mettack (Zügner and Günnemann 2019a) at the early-warning regime $k \in \{1, \ldots, 5\}$ ($\mathbf{149/150}$ paired wins, $p<10^{-43}$).

D.3 Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\max} \in [0.30, 0.99]$ on Cora (10 seeds). Even when the spectral-normalisation cap is pushed to 0.99, driving the theoretical $\varepsilon_{\text{crit}}$ down 230$\times$, the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent grows only $\mathbf{1.17 \to 1.80}$. This is a $\mathbf{2-4}\times$ spectral-radius margin to criticality (ρ=1), the operative quantity behind $\varepsilon_{\text{crit}}$, stable across the sweep. Training converges throughout, and the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\max} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 7 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at N=200 (per-edge τ=0.999), and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (consistent with the trained-IGNN range $\kappa \in [0.14, 0.59]$ of table 2). For larger N the rSVD error (Halko, Martinsson, and Tropp 2011) is bounded by the

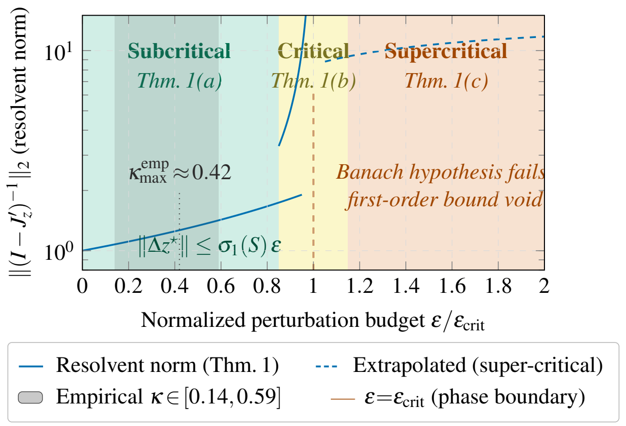

Figure 6: Three-regime characterisation of Theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \to \varepsilon_{\text{crit}}$.

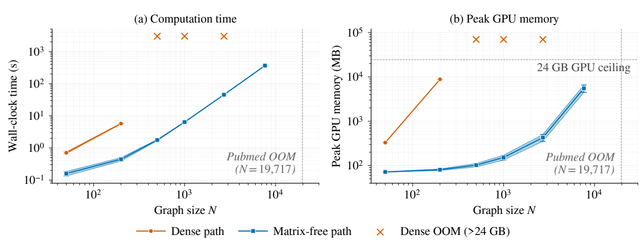

Figure 7: Dense vs. matrix-free AEGISpipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond N=200; matrix-free is roughly $O(N \log N)$ in time, sublinear in memory, reaching Amazon Photo (N=7,650) at 365 s/5.5 GB.

spectral gap; Pubmed (N=19,717) exceeds 24 GB.

D.4 Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at N=200; we use N=50 as default (66× faster). **Full-graph scale.** A 50-node BFS covers only $\sim$1.8% of a citation graph's edges (Cora τ=0.16), so at this scale we run AEGISon the full graph; there its edge advantage over degree-proportional ranking *amplifies* to $\mathbf{9.82}\times$ (Citeseer) and $\mathbf{3.25}\times$ (Cora) at k=10, against $\approx$1.1× on subgraphs. **Hyperparameters.** Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier (r_v=0.147@72.1% vs. 0.089@80.6%).

Edge protection and the delocalization of vulnerability. On the *full* graph (Cora, N=2,708, 10 seeds), masking the top-k edges by v_{ij} cuts $\sigma_1(S_c)$ damage by 2.4±1.8% at k=5 and 4.6±2.9% at k=10 — **12–46**× over random masking, but far below the 42–61% a 50-node subgraph reports (where k=5 is already 10% of its edges). The reason is structural: the leading attack mode is *delocalized* (participation ratio 41–89 edges), so removing a handful excises a negligible slice; reaching subgraph-scale reduction needs 2–8% of edges masked, and an adaptive recomputation erodes the gain a further 1–2 pp. AEGISthus *localizes* risk for attribution and triage rather than yielding a few-edge defense, the failure mode sober-look critiques (Mujkanovic et al. 2022) flag.

Table 5: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall of the *unweighted* v_{ij} ranking vs. single-edge removal; the edge-weighted $A_{ij}v_{ij}$ ranking of Proposition 3 (fig. 3) is the higher headline score. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}\times$	$+.32_{\pm.04}$	$77.5_{\pm1.7}\%$
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm0.6}\times$	$-.04_{\pm.03}$	$78.9_{\pm0.9}\%$
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm0.4}\times$	$+.49_{\pm.02}$	$78.3_{\pm1.8}\%$
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm0.1}\times$	$+.33_{\pm.03}$	$76.6_{\pm1.9}\%$
GAT†	$1.01_{\pm.01}$	$2.1_{\pm0.1}\times$	$+\mathbf{.54}_{\pm.06}$	$77.8_{\pm1.2}\%$
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm0.2}\times$	$+.22_{\pm.10}$	$77.5_{\pm1.5}\%$
APPNP	$1.02_{\pm.00}$	$3.4_{\pm0.3}\times$	$+.35_{\pm.01}$	$\mathbf{82.2}_{\pm0.6}\%$

D.5 Per-Architecture Characterisation (Explicit GNNs)

For K-layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT† (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form S_K=$\partial\text{vec}(Z_K)/\partial\text{vec}(A)$ via finite differences and build S_c from S_K identically (Proposition 4). Table 5 reports the per-architecture characterisation on Cora: IGNN carries Theorem 1; the explicit models receive the computational tool without the regime characterisation.

GAT† scope. Standard GAT uses A as a binary mask, so $\partial Z/\partial A_{ij}$=0 on existing edges and S_c is undefined; our GAT† modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij}\alpha_{ij}Wh_j$), restoring differentiability and yielding tightness 0.99, AtkAdv 4.4×, $\tau = +0.36$. Binary-mask architectures (hard attention, max/min aggregation, GATv2 (Brody, Alon, and Yahav 2022)-style dynamic-attention masking) fall outside the framework. **Robust backbones.** Under the $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite (Zhu et al. 2019) and GNNGuard-lite (Zhang and Zitnik 2020) match or exceed IGNN; the cap is a precondition (without it κ>1 voids Theorem 1). **Cross-dataset transfer detail.** The edge-weighted $A_{ij}v_{ij}$ ranking beats a pure edge-weight baseline by $\Delta\tau \approx +0.25$, rising to +0.65 on the fraud graph where uniform weights are uninformative, so v_{ij} adds rank information beyond the weight. GAT-2† is the exception: attention reweights edges, so the unweighted v_{ij} transfers better there (+0.47 vs. +0.35). Proposition 3's sufficient pairwise condition holds for **47–62**% of edge pairs, so the global τ is empirical, not implied; where unweighted v_{ij} is uninformative (GCN-2's near-uniform 2-hop shifts),

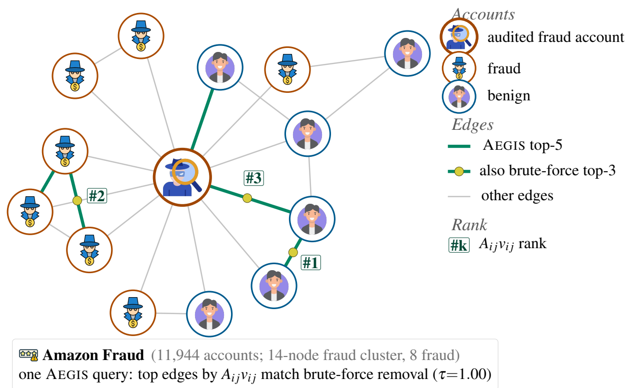

Figure 8: AEGISauditing an IGNN fraud detector on an Amazon Fraud cluster (Dou et al. 2020). Icons mark fraud / benign accounts; the audited account sits at the hub. Green edges are the top-5 adversarial fault lines by the edge-weighted score $A_{ij}v_{ij}$ of Proposition 3; yellow dots mark the three also in the brute-force single-edge-removal damage top-3. The one-query ranking reproduces the brute-force ranking (τ=1.0 on this cluster), at one query's cost rather than $|E|$ reconvergences.

the weighted ranking recovers the order via the edge weight ($-0.28 \rightarrow +0.99$, Citeseer).

E Fraud-Detector Audit: Walk-Through

This appendix expands the audit-in-action summarised in section 6. GNN fraud detectors are a prime adversarial target: a flagged account can try to evade detection by perturbing a few of its behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGISaudits such a detector directly — no labels, no retraining — reading from one S_c query which edges most threaten its predictions.

Figure 8 shows the audit: the trained IGNN flags the central account, and the edge-weighted ranking $A_{ij}v_{ij}$ pinpoints the fault lines, reproducing the brute-force single-edge-removal damage ranking from a single S_c query. This auditing claim is what the paper validates at scale, and *non-circularly*. The edge-weighted ranking transfers across the full fraud graph (fig. 3), precisely where uniform edge weights are uninformative so v_{ij} adds the most rank information ($\Delta\tau$ up to $+\mathbf{0.65}$ over the weight baseline). The SVD direction transfers from a *separately trained* surrogate (zero shared gradients, $\cos = 0.99$; section 5.2), so v_{ij} reflects model-intrinsic vulnerability rather than a gradient artifact; AEGISout-damages discrete attackers (Mettack, GR-BCD) on their own metric in the early-warning regime (sections D.2 and 5.1); and the per-edge v_{ij} scores prioritise which connections to monitor, though adversarial vulnerability itself proves delocalized at scale (section D.4). Beyond a smoothing certifier, S_c also names *which* edges and *in which direction* (section 3.1), turning a one-query audit into an actionable defense.

F Reproducibility

All results use 10 fixed random seeds $\{42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999\}$; reported $\pm$ are standard deviations over these seeds. Models are trained in PyTorch (Paszke et al. 2019); IGNN (Gu et al. 2020) uses $F(Z)=\mathrm{ReLU}(\hat{A}ZW^\top + U(X))$ with $W \in \mathbb{R}^{64\times 64}$ spectral-normalised (Miyato et al. 2018) and Adam (lr 0.01). Datasets are the public splits of Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), and Amazon Fraud (Dou et al. 2020). The randomized SVD (Halko, Martinsson, and Tropp 2011) uses $k=p=10$, $n_{\text{iter}}=2$; the Neumann series early-stops at $\left\|J_z^k b\right\| < 10^{-6}\,\|b\|$. Timings are on a single RTX 4090. Code release follows the disclosure protocol of section 8: the diagnostic-only path (r_v and v_{ij} scores) is released unconditionally, while the attack-direction reconstruction (algorithm 1, steps 3–4) is gated behind institutional-affiliation review.